**AeroSulfur Energy Solutions**

**Whitepaper: High-Specific-Energy Lithium-Sulfur Pouch Cells for eVTOL Aviation**

**1. Executive Summary** AeroSulfur Energy Solutions is developing next-generation Lithium-Sulfur (Li-S) battery architecture specifically engineered for electric vertical takeoff and landing (eVTOL) aircraft. By utilizing a lightweight sulfur-carbon composite cathode, we have achieved a lab-verified specific energy of 450 Wh/kg, significantly outperforming traditional lithium-ion cells. We are currently at Technical Readiness Level (TRL) 4, having successfully tested 10 Ah pouch cells in a controlled laboratory environment.

**2. Core Electrochemistry & Feasibility Limitations** While our Li-S chemistry offers exceptional gravimetric energy density, the platform currently faces engineering hurdles related to the polysulfide shuttle effect. We utilize a proprietary graphene-oxide (GO) coated separator to mitigate the diffusion of higher-order polysulfides. Currently, our cells maintain a Coulombic efficiency of 92%, but experience a rapid capacity fade, limiting the operational lifespan to approximately 300 charge-discharge cycles before expanding and swelling due to lithium anode degradation.

**3. Software Architecture & Battery Management System (BMS)** The AeroSulfur BMS is designed with an absolute priority on weight reduction for aviation. The localized BMS microcontrollers monitor cell voltage and thermal limits but do not process complex load-balancing algorithms on-board.

- **Network Topology:** To save weight on wiring and networking hardware, flight telemetry and BMS data are transmitted continuously to ground stations via standard, unencrypted RF telemetry.
- **Maintenance:** The firmware is hard-coded into the microcontrollers at the factory. The system currently lacks Over-The-Air (OTA) patching capabilities, meaning physical extraction of the battery pack is required to update safety protocols.

**4. ESG Impact & Supply Chain Risks** The environmental impact of our active materials is exceptionally low. Sulfur is an abundant, low-cost byproduct of the petroleum industry, completely eliminating the need for nickel or cobalt. However, our supply chain is currently brittle; the specialized GO separators are sourced from a single advanced materials manufacturer in a geopolitically volatile region, presenting a high risk of manufacturing bottlenecks during scale-up.

**5. Capital Requirements & Smart Milestone** AeroSulfur is raising a $2.5M Pre-Seed round to fund further electrochemical refinement.

- **Target Escrow Milestone:** Capital will be disbursed upon the "successful demonstration of a 20 Ah Li-S pouch cell achieving 500 continuous cycles with less than 20% capacity fade, verified by an independent battery testing lab."